

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

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Research & Professional Profile

Applied multimodal intelligence researcher and Senior Data Scientist at Nasdaq; Ph.D. in Computer Science. Expertise in generative vision, efficient adaptation, financial ML, and cloud-based ML systems.

Professional Experience

Nasdaq

St. John's, NL, Canada

Senior Data Scientist, AI/Analytics

Mar. 2025 – Present

- Integrated an **existing cross-team ML UI/backend prototype** into a **SageMaker workspace**; configured networking/IAM and **PostgreSQL user-state persistence**; deployed **FastAPI on AWS Lambda using Mangum**.
- Designed cross-scale ensemble feature selection that removed **77.2% of input features** while retaining competitive performance on CFML data; presented the algorithm to more than 60 attendees.
- Built **SageMaker Job Submitter** for script/dependency packaging across **Processing, Training, Hyperparameter Tuning, and SageMaker Pipelines**; cut developer job configuration from **7 days to 1 day (about 86% less)**.
- Designed **feature engineering with PyTorch/CUDA-accelerated feature-label AUROC screening**; used **Amazon Bedrock** to assess candidate features' semantic plausibility.
- Developed **PagerDuty RAG assistant v2** on the company's GenAI platform; **now used internally and recognized in a company shout-out**. Retrieves incident context and troubleshooting guidance for on-call engineers.
- Mentored **cheque OCR** research on perceptually aligned quality metrics, culminating in a student presentation; guided **tabular-data generation** leading to a **StDDPM manuscript submission**.
- Mentored **Seeam (work-term student)** on an **input payload checker** from Jun. 2026, and **Daniel (Mitacs student)** on **GNN-based crime-ring detection** from Sep. 2026.

AI Algorithm Intern (two placements)

May 2024 – Jan. 2025

- Proposed **TextDoctor**: structure-pyramid prediction and patch-pyramid diffusion for document/cheque inpainting; **8K inpainting on a 24GB RTX 3090**. Python/PyTorch, AWS/SLURM/CUDA. **First-author manuscript submitted to Engineering Applications of Artificial Intelligence (EAAI)**.
- Evaluated **7 document datasets** without dataset-specific fine-tuning. FUNSD, TextDoctor (GSDM): **55.38% PaddleOCR word accuracy vs. 45.24% DocDiff (+10.14 percentage points)**.
- Designed **PLAID**: patch-level Gaussian discriminant modeling of genuine and forged samples, **without backbone fine-tuning**; evaluated **8 biometric benchmarks and 16 backbones**. **8-shot ResNet-152 AUROC: 99.42% CEDAR, 98.42% GPDS-150**.
- **First-author submission to IEEE Transactions on Information Forensics and Security (TIFS)**; major revision submitted **Apr. 2026**, awaiting second-round decision. Presented both placements: earlier talk about **40–60 attendees**; final talk **39 cross-functional attendees**.

AI Intern – Beaufort Solutions

Jul. 2022 – Nov. 2022

St. John's, NL, Canada

- Led method design of **TEG**, a CLIP-based **few-shot adaptation** approach for personalized photo-book curation; introduced a text-embedding-guided classifier and auxiliary classification loss to improve learning from limited labels.
- Led Theme25 collection pipeline (**35,655 annotated images, 25 themes**); evaluated Theme25, CIFAR100, and ImageNet on AWS/SLURM; published in **Journal of Electronic Imaging (2024)**.

AI Algorithm Intern – Zhongkong Hanlian

Dec. 2017 – Aug. 2018

Hangzhou, China

- Designed **ResNet–DenseNet fusion** for vehicle-logo recognition; built Java annotation tools and a dataset for training and evaluation; **deployed with Caffe for real-time inference at community security gates**.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada	2021–2025
M.Eng., Computer Software and Theory – Wenzhou University, China	2018–2021
B.Eng., Digital Media Technology – Communication University of Zhejiang, China	2014–2018

Selected Research & Engineering Projects

UltraDiffEdit – High-Resolution Generative Editing [Code] TNNLS 2026; first author

- Developed multi-patch encoding, global-local consistency denoising, and progressive sampling to extend pretrained latent diffusion models to real-image editing without fine-tuning.
- Demonstrated **8K editing on one RTX 3090**; achieved **UHRSDEdit (4K–8K) crop-FID 7.45 vs. 18.76 for SDXL+bicubic (60.3% lower)**. Video.
- Built **three benchmarks totaling 1,188 images**: DIV2KEdit, Syn2KEdit, and UHRSDEdit; evaluated pretrained SDXL and ControlNet without fine-tuning.

FACEMUG – Multimodal Local Facial Editing TVCG 2024; first author

- Developed a unified editing framework for text, sketches, semantic maps, color maps, exemplars, and attribute labels, with multimodal feature fusion and self-supervised latent warping.
- Achieved **FFHQ FID 23.16 vs. 52.39 for Unite&Conquer** under sketch/semantic conditioning; received **77.88% preference** in a **52-participant CelebA-HQ study** with sketch/semantic/text guidance. Publication.

AdaMSS – Parameter-Efficient Adaptation [Code] NeurIPS 2025; co-author

- Co-developed adaptive multi-subspace updates; ViT-Large averaged **90.13% vs. LoRA's 85.40% (+4.73 percentage points)** across seven tasks, with **241K vs. 1.57M trainable parameters (15.4%)**.
- LLaMA 2-7B reached **50.80% GSM8K accuracy with 4M trainable parameters** vs. 49.05% with 6,738M for full fine-tuning. Method **integrated into Hugging Face PEFT**.

DocPure – Unified Document Restoration [Code] TCSVT 2026; co-first author

- Co-developed degradation-aware structure extraction and wavelet modulation; tested **one prompt-free model on 12 datasets** for deblurring, denoising, compression artifacts, and deshadowing.
- On TDD deblurring, achieved **32.55 dB PSNR vs. DocRes's 28.34 dB (+4.21 dB)** and **61.27% PaddleOCR word accuracy vs. 55.66% (+5.61 percentage points)**. Benchmarks.

GRIG – Data-Efficient Image Inpainting [Code] CVM 2025; first author

- Designed CNN–Transformer iterative residual refinement and forgery-patch adversarial training for data-efficient inpainting on **10 few-shot datasets with 18–750 training images**.
- Reduced FID by **3.70–26.48% vs. each dataset's next-best baseline at 50–60% masking**; benchmarks span medical, animal, cartoon, and art images. Paper.

Visual Style Prompt Learning [Code] Pattern Recognition 2025; first author

- Developed latent-diffusion style prompts and style-modulated aggregation with **four denoising steps**; achieved **12.3 FPS at 512 × 512 on an RTX 3080 Ti**.
- Achieved **WebPhoto-Test FID 74.25 vs. CodeFormer's 83.17 (10.7% lower)**; best mean rank (3.67) across FID/NIQE on three real-world test sets. Paper.
- On **CelebA-Test**, reduced **FAN landmark NME from 6.08% to 2.43%** and raised **HSEmotion agreement from 79.73% to 86.03%**; reference landmarks and emotion labels are model predictions on clean images.

Local Agentic Code Editor Nasdaq internal prototype; 2026

Exploratory side project: API-driven code editing and local tool orchestration

- Built a working local AI code-editor prototype with **model APIs and agentic routing** across code generation, review and shell-tool execution.
- Implemented an iterative debugging loop that **diagnoses bugs, edits code, applies fixes and uses execution feedback** to determine the next action.

Peer-Reviewed Publications

Tuning-Free Latent Diffusion Models for Ultrahigh-Resolution Image Editing

Wanglong Lu, Lingming Su, Kaijie Shi, Minglun Gong, Xiaogang Jin, Hanli Zhao, Xianta Jiang

IEEE Transactions on Neural Networks and Learning Systems, 1-15, 2026

UHDRes: Ultra-High-Definition Image Restoration via Dual-Domain Decoupled Spectral Modulation

Shihao Zhao, Wanglong Lu, Binhao Wang, Tao Wang, Kaihao Zhang, Hanli Zhao

IEEE Transactions on Circuits and Systems for Video Technology, 2026

DocPure: Prompt-Free Unified Document Restoration via Degradation-Aware Structure-Guided Wavelet Modulation

Lingming Su, Wanglong Lu, Tao Wang, Kaihao Zhang, Nan Zhang, Liyan An, Hanli Zhao

IEEE Transactions on Circuits and Systems for Video Technology, 2026

EchoSR: Efficient Context Harnessing for Lightweight Image Super-Resolution

Hanli Zhao, Binhao Wang, Shihao Zhao, Tao Wang, Kaihao Zhang, Wanglong Lu

Information Fusion, 104471, 2026

AdaMSS: Adaptive Multi-Subspace Approach for Parameter-Efficient Fine-Tuning

Jingjing Zheng, Wanglong Lu, Yiming Dong, Chaojie Ji, Yankai Cao, Zhouchen Lin

NeurIPS 2025

Visual style prompt learning using diffusion models for blind face restoration

Wanglong Lu, Jikai Wang, Tao Wang, Kaihao Zhang, Xianta Jiang, Hanli Zhao

Pattern Recognition, 2025

Toward Biosignals-Free Autonomous Prosthetic Hand Control via Imitation Learning

Kaijie Shi, Wanglong Lu, Hanli Zhao, Vinicius Prado da Fonseca, Ting Zou, Xianta Jiang

IEEE Transactions on Neural Systems and Rehabilitation Engineering 33, 3544-3554, 2025

GRIG: Data-Efficient Generative Residual Image Inpainting

Wanglong Lu, Xianta Jiang, Xiaogang Jin, Yong-Liang Yang, Minglun Gong, Kaijie Shi, Tao Wang, Hanli Zhao

Computational Visual Media 11(6), 1329-1361, 2025

Do Inpainting Yourself: Generative Facial Inpainting Guided by Exemplars

Wanglong Lu, Hanli Zhao, Xianta Jiang, Xiaogang Jin, Yongliang Yang, Min Wang, Jiankai Lyu, and Kaijie Shi

Neurocomputing 617, 128996, 2025

Degradation-Aware Frequency-Separated Transformer for Blind Super-Resolution

Hanli Zhao, Binhao Wang, Wanglong Lu, Juncong Lin

International Conference on Computational Visual Media, 2025

Real-time dual-eye collaborative eyeblink detection with contrastive learning

Hanli Zhao, Yu Wang, Wanglong Lu, Zili Yi, Jun Liu, Minglun Gong

Pattern Recognition, 2025

FACEMUG: A Multimodal Generative and Fusion Framework for Local Facial Editing

Wanglong Lu, Jikai Wang, Xiaogang Jin, Xianta Jiang, and Hanli Zhao

IEEE Transactions on Visualization and Computer Graphics, 2024

Handling the Non-smooth Challenge in Tensor SVD: A Multi-objective Tensor Recovery Framework

Jingjing Zheng, Wanglong Lu, Wenzhe Wang, Yankai Cao, Xiaoqin Zhang, and Xianta Jiang

ECCV, 2024

Restoring Vision in Hazy Weather with Hierarchical Contrastive Learning

Tao Wang, Guangpin Tao, Wanglong Lu, Kaihao Zhang, Wenhan Luo, Xiaoqin Zhang, Tong Lu.

Pattern Recognition Volume 145, January 2024, 109956

TEG: image theme recognition using text-embedding-guided few-shot adaptation

Jikai Wang, Wanglong Lu, Yu Wang, Kaijie Shi, Xianta Jiang, Hanli Zhao

Journal of Electronic Imaging 33(1), 013028, 2024

Detecting Blinks from Wearable Cameras using Spatial-Temporal-Aware Deep Network Learning

Yu Wang, **Wanglong Lu**, Hanli Zhao, Xianta Jiang, Bin Zheng, M. Stella Atkins.

ETRA 23, 2023

Efficient Generative Face Completion with Perceptual Deblurring

Hanli Zhao, Ying Liu, **Wanglong Lu**, Xiaogang Jin, Hui Huang, Kaijie Shi

Journal of Computer-Aided Design & Computer Graphics 34(9), 1420-1431, 2022

Generative image inpainting with enhanced gated convolution and Transformers

Min Wang, **Wanglong Lu**, Jiankai Lyu, Kaijie Shi, Hanli Zhao.

Displays 76, 102321, 2022

Probability-Based Channel Pruning for Depthwise Separable Convolutional Networks

Han-Li Zhao, Kai-Jie Shi, Xiaogang Jin, Ming-Liang Xu, Hui Huang, **Wanglong Lu**, Ying Liu.

Journal of Computer Science and Technology 37, 584-600, 2022

Image Splicing Tamper Detection Based on Multi-Scale Feature Priors [in Chinese; translated title]

Wanglong Lu et al.

China Sciencepaper 17(11), 2022

Category-consistent deep network learning for accurate vehicle logo recognition

Wanglong Lu, Hanli Zhao, Qi He, Hui Huang, Xiaogang Jin.

Neurocomputing 463, 623-636, 2021

High-quality retinal vessel segmentation using generative adversarial network with a large receptive field

Hanli Zhao, Xiaqing Qiu, **Wanglong Lu**, Hui Huang, Xiaogang Jin.

International Journal of Imaging Systems and Technology 30(3), 828-842, 2020

Preprints

Simulation-Driven Imitation Learning for Biosignals-Free Shared-Autonomy Prosthetic Grasping

Kaijie Shi, **Wanglong Lu**, Han Chen, Vinicius Prado da Fonseca, Ting Zou, Hanli Zhao, Xianta Jiang

arXiv preprint arXiv:2606.07389, 2026

TextDoctor: Unified Document Image Inpainting via Patch Pyramid Diffusion Models

Wanglong Lu, Lingming Su, Jingjing Zheng, Vinícius Veloso de Melo, Farzaneh Shoeleh, John Hawkin, Terrence Tricco,

Hanli Zhao, Xianta Jiang

ArXiv, 2025

StDDPM: Sequential Financial Synthetic Data Generation using T-distribution based Diffusion Model

Omid Tarkhaneh, Hamideh Mehri, **Wanglong Lu**, Farzaneh Shoeleh, Vinicius Veloso de Melo

Research Square preprint, 2025

Promptrr: Diffusion models as prompt generators for single image reflection removal

Tao Wang, **Wanglong Lu**, Kaihao Zhang, Wenhan Luo, Tae-Kyun Kim, Tong Lu, Hongdong Li, Ming-Hsuan Yang

ArXiv, 2024

Dissertation

Generative Models for Semantic Facial Image Editing: Multimodal Approaches

Wanglong Lu

PhD thesis, Memorial University of Newfoundland, 2025

Technical Skills

Programming: Python, PyTorch, SQL, Git, Bash, C++, Java, CUDA, OpenCV.

ML & GenAI: Latent diffusion, GANs, Transformers, multimodal conditioning, vision-language models, PEFT/LoRA, Hugging Face, RAG, feature selection, experimental evaluation.

Infrastructure: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, SLURM.

Additional Engineering Project

AI Headshot Generation: Implemented model invocation and deployed the generation application.

Teaching, Mentorship & Talks

- Mentored **12 students** in generative vision, image restoration/recognition, and applied ML research.
- Cheque OCR mentorship identified promising image-quality metrics consistent with human perception and supported text-extraction evaluation; the student completed a research presentation.
- Guided sequential financial-data generation research leading to **StDDPM**, a submitted manuscript with a Research Square preprint; evaluated on **1.06 million Czech banking transactions across 4,500 accounts**.
- Teaching assistant at Memorial: Data Structures & Algorithms (2021); Introduction to Machine Learning (2024).
- Guest lecturer, Bio-Inspired Robotics (ENGI 9986), Memorial University, Feb. 2025: a 150-minute lecture on Deep Generative Models and Applications for 10 graduate students.
- Invited speaker, IEEE Newfoundland and Labrador Section AGM, Mar. 2026: Generative Models for Semantic Image Editing: Multimodal Approaches.
- Research talks for academic/industry audiences; 2026 feature-selection presentation to **60+ attendees**.

Academic Service

Invited reviewer, 2026: IEEE TPAMI, IEEE TMM, IEEE Journal of Biomedical and Health Informatics (JBHI), Pattern Recognition, Applied Soft Computing, Engineering Applications of Artificial Intelligence, Scientific Reports, Neurocomputing, Knowledge-Based Systems, and ECCV.

2025: IEEE TIP, IEEE TMM, Pattern Recognition, Neurocomputing, CAD/Graphics, Advances in Manufacturing, Signal, Image and Video Processing, IEEE SPL, Multimedia Systems, Journal of Machine Learning and Cybernetics.

2024: IEEE TMM, IEEE TCSVT, Knowledge-Based Systems, Journal of Medical Systems, Cluster Computing, Journal of Visual Communication and Image Representation, Displays, The Visual Computer, and ETRA PETMEI.

Granted Patents

Co-inventor on four Chinese invention patents:

- Category-consistent deep network learning for image recognition. ZL 2021 1 0408724.X.
- Deep feature-based convolutional neural network for vehicle-logo recognition. ZL 2020 1 0139043.3.
- GAN-based intelligent segmentation of retinal blood-vessel images. ZL 2019 1 0884346.5.
- Convolutional neural network-based intelligent vehicle-logo detection. ZL 2020 1 0139068.3.

Selected Honors & Awards

- Computer Innovation and Entrepreneurship Award, Wenzhou Computer Federation; first prize, team leader (2024).
- Genesis entrepreneur training program, Memorial University (2024).
- Outstanding Research Award (Ph.D.), Computer Science, Memorial University (2023).
- Mitacs Accelerate Award (2022).
- Outstanding Graduate, Wenzhou University (2020); Outstanding Graduate of Zhejiang Province (2018).
- Chinese Graduate Mathematical Modeling Competition, third prize (2019); Lanqiao Cup C/C++ Programming Competition, second prize (2019); CCPC regional competition, third prize (2019).